Paleomagnetism of Mesozoic plutons in the westernmost Coast Complex of British Columbia

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The Lower Cretaceous Stephens Island (102 ± 8 Ma) and Captain Cove (109 ± 6 Ma) plutons and the Upper Jurassic Gil Island (136 ± 3 Ma) and Banks Island (144 ± 6 Ma) plutons belong to the western K–Ar age zone of the N 35° W trending Coast plutonic complex southwest of Prince Rupert, B.C. After removal of initial viscous components, AF demagnetization isolates a stable primary remanence at 36 of 49 sites (10 specimens from 5 cores/site) before anhysteretic components are added. All sites have normal polarity which is consistent because their K–Ar ages fall in the predominantly normal Cretaceous and Jurassic Quiet Intervals. The poles for Stephens Island (339° W, 67° N (7°, 10°)), Captain Cove (9° W, 72° N (8°, 11°)), and Gil Island (357° N, 70° N (6°, 8°)) lie just north of Britain and are discordant for the North American craton. The tectonic panel including these plutons was tilted ≃ 30° SW during the Upper Cretaceous – Paleocene orogeny as the leading edge of the North American plate overrode the subducting oceanic Kula Plate. This interpretation is supported by other arguments including the attitudes of contacts and foliations, plutonic trend directions, distribution of metamorphic grades, and paleomagnetic data from the area to the east. The Banks Island pluton lies in the tectonic panel to the west. Its pole of 210° W, 81° N (33°, 38°) is poorly defined but apparently concordant.

Les plutons de Stephens Island (103 ± 8 Ma) et de Captain Cove (109 ± 6 Ma) du Crétacé inférieur et les plutons de Gil Island (136 ± 3 Ma) et de Banks Island (144 ± 6 Ma) du Jurassique supérieur font partie de la zone occidentale d'âge K–Ar du complexe plutonique Côtier, orienté selon N 35° O au sud-ouest de Prince-Rupert, C.-B. Après suppression des composantes visqueuses initiales, la désaimantation AF isole une rémanence primaire stable à 36 des 49 sites (10 spécimens provenant de 5 carottes par site) avant que les composantes anhystérétiques soient ajoutées. Tous les sites ont une polarité normale, ce qui s'explique par leur âge K–Ar qui tombe dans les intervalles calmes de polarité normale au Crétacé et au Jurassique. Les pôles pour Stephens Island (339° O, 67° N (7°, 10°)), Captain Cove (9° O, 72° N (8°, 11°)) et Gil Island (357° O, 70° N (6°, 8°)) se situent juste au nord de la Grande-Bretagne et sont discordants pour le craton de l'Amérique du Nord.

Le panneau tectonique comprenant ces plutons a été incliné d'environ 30° au sud-ouest durant l'orogénèse du Crétacé supérieur-Paléocène, alors que la bordure de la plaque de l'Amérique du Nord chevauchait la plaque océanique Kula en subduction. Cette interprétation est supportée par d'autres observations comme les attitudes des contacts et des foliations, les directions d'orientation des plutons, la distribution des grades de métamorphisme et les données paléomagnétiques pour la région à l'est. Le pluton de Banks Island se trouve dans le panneau tectonique à l'ouest. Son pôle (210° O, 81° N (33°, 38°)) est mal défini mais semble concordant.

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Introduction

The Coast Mountains of British Columbia are underlain by one of the world's largest Phanerozoic batholiths. Hutchison (1967) noted that the plutons within the complex could be divided into three K–Ar radiometric age zones which parallel the north-northwest trend of the mountains. These are the western (84–140 Ma[1]), central (64–80 Ma), and eastern (40–50 Ma) zones (Hutchison 1970; Monger and Hutchison 1971). Lacking critical geologic field relations, Hutchison (1967) considered three possible hypotheses to explain the zones: (i) they date the time of sequential plutonic emplacement and primary cooling; (ii) they date the passage of a thermal metamorphic front from west to east; or (iii) they date progressive uplift and cooling from west to east. In two paleomagnetic studies on plutons in the central and eastern zones, Symons (1974*a*) found that their remanence polarity pattern and directions were incompatible with both the thermal metamorphic and uplift hypotheses,

[1]m.y. = Ma in SI units.

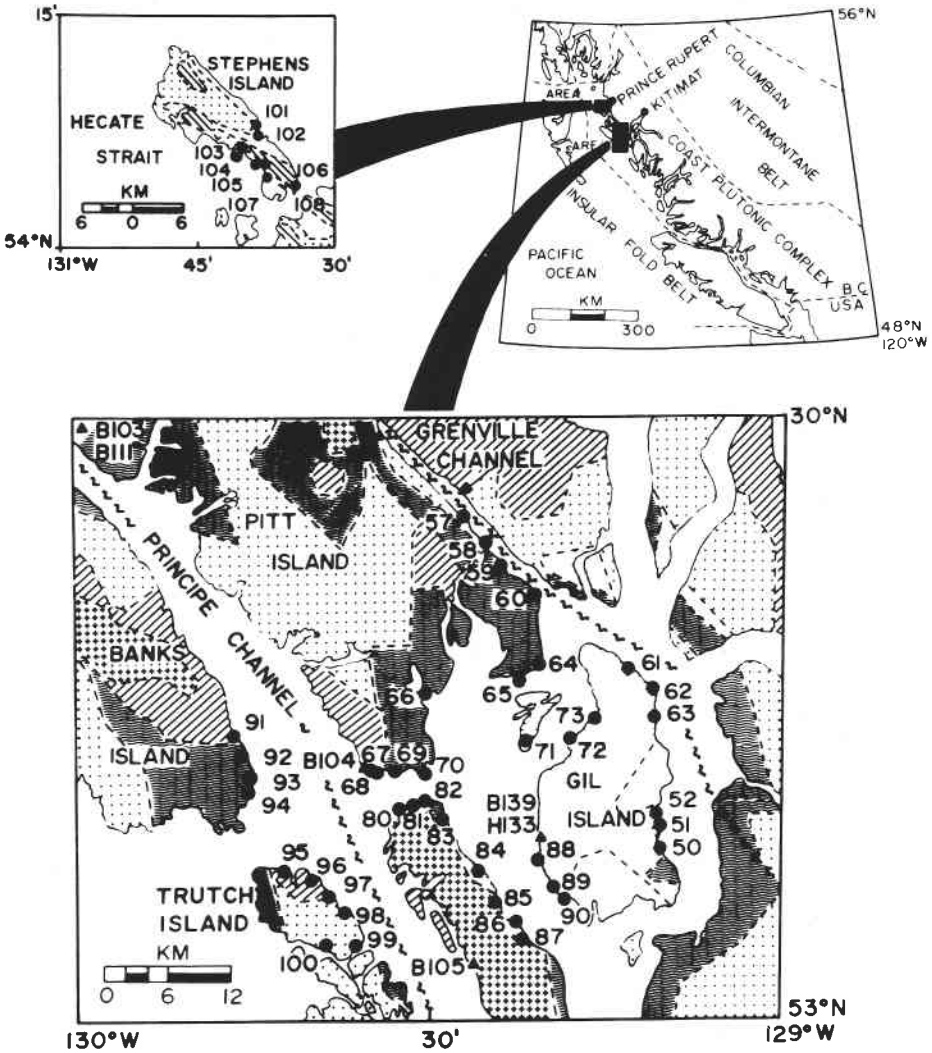


FIG. 1. Sampling site locations. Geologic patterns: gneissic diorite (horizontal wavy lines), hornblende diorite (horizontal lines), quartz diorite (stipple), granodiorite (diagonal line), and monzonite (crosses).

and further (Symons 1977) that remanence directions predate regional folding of the plutons and must record their initial cooling. Prior to this work there had been no substantial paleomagnetic study of rocks in the western zone. In this paper paleomagnetic data from four plutons in the western zone, outcropping west and southwest of Prince Rupert (Fig. 1) are interpreted in geochronologic and geotectonic terms.

Geology

The four plutons sampled were the Stephens Island pluton in the Prince Rupert – Skeena map-area (Hutchison 1967) (Fig. 1) and the Captain Cove, Gil Island, and Banks Island plutons in the Douglas Channel – Hecate Strait map-area (Roddick 1970). The plutons intrude limited areas of metasedimentary and metavolcanic rocks of upper Paleozoic (Permian or older?) to Middle Jurassic age (Hutchison 1970). These rocks exhibit Barrovian-type metamorphism and now comprise schists, gneisses, and migmatites of the greenschist and hornblende amphibolite facies.

The western zone is bounded on the west by Hecate Strait which is apparently underlain by Triassic rocks of the Insular Belt. The main evidence is that flat-lying Karmutsen volcanics outcrop on Bonilla Island some 8 km west of Banks Island which give a K–Ar whole rock age of 202 ±21 Ma (GSC 72–2 in Wanless *et al.* 1973). The western zone is bounded on the east by the Grenville Channel fault across which outcrop rocks of the 64 to 80 Ma central age zone. The western zone is itself split by the Principe Laredo fault along Principe Channel and Estevan Sound into western and eastern panels.

The western panel is represented by the large 100 by 20 km Banks Island pluton outcropping on Banks, Trutch, and the Estevan Islands (Fig. 1). The eastern panel includes the northernmost Dundas Island pluton. South of it is the small 30 by 6 km Stephens Island pluton, followed by the Porcher Island pluton, and then by the large 100 by 25 km Captain Cove pluton outcropping on McCauley, Pitt, and Campania (?) Islands. All of these plutons have massive monzonite and granodiorite cores, surrounded by a quartz diorite phase and then by a peripheral gneissic diorite–gabbro–migmatite phase. At the south end of the Captain Cove pluton occurs the largest basic complex of the Coast Mountains. It is the circular Gil Island diorite comprised of a massive undifferentiated hornblende diorite core some 20 km across surrounded by a 5 km wide peripheral gneissic diorite–gabbro–migmatite phase.

Several biotite, hornblende, and whole-rock K–Ar ages are available from plutonic rocks in the western zone (Wanless *et al.* 1965, 1966, 1967, 1968, 1970, and 1972). In the eastern panel are ages from the Dundas Island pluton of 96 ±5, 101 ±15, and 84 ±4 Ma (GSC 66-4 and 5, and 67-21) and from the Captain Cove pluton of 103 ±6, 111 ±6, 115 ±6, and 104 ±4 Ma (GSC 64-5 and 6, and 67-23 and 24). The age of the intervening Stephens Island pluton is taken as the average of these seven ages. Farther south in the Bella Coola – Laredo Sound map-area (Baer 1967) ages of 87 ±11, 88 ±5, 109 ±5, 100 ±6, 90 ±4, and 80 ±4 Ma come from the plutonic rocks and ages of 87 ±5 and 81 ±4 Ma (GSC 67-29 and 30) from a synmagmatic dike. All of these ages fall in the 80 to 115 Ma range. The Gil Island complex is older than the rest of the eastern panel by at least 25 Ma giving ages of 139 ±7 and 133 ±22 Ma (GSC 66-10 and 11). The plutonic rocks in the western panel appear to be even older giving an age of 144 ±6 Ma (R. K. Wanless, personal communication, 1976).

The rocks of the western zone have been intruded or overlain by syenogranitic plutons giving ages of 12 ±2 and 13 ±2 Ma (GSC 70-8 and 9), by undeformed rhyolitic flows and related blue porphyritic dikes of 14.5 ±1.0 Ma (GSC 67-31), and by brown basaltic flows and dikes of 12.5 ±2.7 Ma (GSC 67-32). The flows and dikes give paleomagnetic poles that are concordant with other Miocene poles from rocks in the North American craton (Symons 1969).

The core of the Coast Range Crystalline Belt has been subjected to considerable uplift. Evidence for uplift comes from the metamorphic grade of the host rocks. Going from west to east, the Triassic rocks on Bonilla Island in the Insular Belt are flat-lying and unmetamorphosed or slightly metamorphosed. The amount of metamorphism increases progressively exposing chlorite grade in the western zone (Hutchison 1967; Roddick 1970), amphibolite grade in the central zone, and granulite grade on the western side of the eastern zone. The amphibolite and granulite mineral assemblages suggest 20 ±5 km of uplift (Hollister 1975, 1976). From this maximum of the Skeena Arch or Foch Antiform, the grade diminishes eastwards through the eastern zone to unmetamorphosed host rocks at the margin with the Interior Plateau (Duffel and Souther 1964; Eisbacher 1973).

Experimental Methods

The 49 sampling sites (Fig. 1) were selected to represent as many plutonic phases as possible (Table 1) and to be several kilometres apart. At each site five cores were drilled several metres apart and oriented *in situ* using a solar compass with Brunton compass and topographic sitings to an accuracy of $<2°$. Two 2.45 cm cylindrical specimens were cut from each core. The specimens' natural remanent magnetizations (NRM) were measured using a Schonstedt SSM-1A spinner magnetometer. Pilot specimens with an average NRM direction and intensity were selected for each site. These were step demagnetized in alternating field (AF) peak intensities of 50, 100, 150, 200, 300, 400, 550, 700, 850, and 1000 Oe (3.98×10^3, 7.96×10^3, 11.94×10^3, 15.92×10^3, 23.88×10^3, 31.60×10^3, 43.78×10^3, 55.72×10^3, 67.66×10^3, and 79.60×10^3

TABLE 1. Sampling and remanence data by sites

Site	Location °W	Location °N	Pluton and phase	AF intens. (Oe)	Data screening a	b	c	N	Remanent magnetization Length (*R*)	Decl.	Incl. (+ down)	α_{95}
50	129.16	53.13	G-7b	550			1	4	3.952	22.5	39.9	12*
51	129.17	53.17	G-7b	400				5	4.950	14.7	63.2	8.6*
52	129.18	53.18	G-7b	550				5	3.986	43.3	58.5	6.4*
57	129.44	53.43	C-8b	200	2			3	2.876	7.7	87.9	32*
58	129.42	53.42	C-8b	200	2			3	2.961	4.9	68.1	18*
59	129.39	53.39	C-5b	200	4	1		0				
60	129.34	53.34	C-5b	200	2			3	2.864	6.4	60.2	33*
61	129.46	53.30	G-7b	700				5	3.403	355.9	44.0	62
62	129.43	53.28	G-7b	300				5	4.937	35.0	11.6	9.7*
63	129.42	53.26	G-7b	300	1		1	3	2.944	357.4	58.7	21*
64	129.33	53.30	C-5b	300				5	4.081	21.9	63.1	42
65	129.37	53.29	C-5b	300				5	4.925	33.2	29.8	11*
66	129.51	52.37	C-8a	200				5	4.936	23.2	66.9	9.8*
67	129.58	53.21	C-9b	200				5	4.760	8.5	70.7	19*
68	129.60	53.22	C-9b	300				5	4.928	27.0	52.6	10*
69	129.55	53.21	C-5b	500				5	4.832	359.3	47.1	16*
70	129.52	53.21	C-8a	300	1			4	3.972	36.0	55.0	9.0*
71	129.36	53.23	G-7b	200	1		1	3	2.774	13.3	56.6	44
72	129.30	53.24	G-7b	550				5	3.611	195.1	86.5	55
73	129.27	53.28	G-7b	400				5	3.394	26.4	71.8	62
80	129.54	53.18	C-9b	200				5	4.365	31.2	47.7	33*
81	129.52	53.19	C-9b	200				5	4.955	356.5	68.9	8.2*
82	129.51	53.19	C-5b	200	3			2	1.989	13.4	54.8	27*
83	129.48	53.18	C-5b	200				5	4.983	13.4	57.7	5.0*
84	129.42	53.14	G-7b	400	1			4	3.990	13.5	55.0	5.2*
85	129.40	53.10	C-10a	200	2			3	2.893	28.5	56.4	29*
86	129.39	53.09	C-10a	200	2			3	2.955	16.2	72.9	19*
87	129.36	53.07	C-10a	200	2			3	2.832	59.8	59.0	38*
88	129.34	53.13	G-7b	400			1	4	3.518	263.6	78.9	40*
89	129.33	53.11	G-7b	550	2			3	2.849	61.9	17.5	35*
90	129.31	53.10	G-7b	550	1			4	3.984	23.3	45.2	6.8*
91	129.78	53.25	B-9b	200	4	1		0				
92	129.77	53.23	B-5b	200				5	4.862	2.2	65.2	15*
93	129.75	53.21	B-5b	200				5	1.665	59.3	33.8	>90
94	129.76	53.19	B-5b	200	4	1		0				
95	129.71	53.13	B-9b	100	2			3	2.799	348.0	82.7	40*
96	129.67	53.12	B-9b	100	2			3	2.685	298.6	40.9	54
97	129.65	53.11	B-8b	100	3			2	1.782	320.2	65.8	>90
98	129.62	53.10	B-8b	200	2			3	2.900	272.8	46.6	28
99	129.61	53.05	B-8b	100	3			2	1.925	5.3	74.0	75
100	129.65	53.05	B-8b	200	2			3	2.857	34.4	63.3	34*
101	130.64	54.13	S-8	100	2			3	2.982	1.0	61.9	12*
102	130.64	54.12	S-8	100	1			4	3.954	27.7	59.9	11*
103	130.65	54.11	S-8	200	2			3	2.992	8.7	49.5	7.7*
104	130.67	54.10	S-9	200	1			4	3.959	16.8	47.3	11*
105	130.65	54.08	S-9	200				5	3.168	325.2	42.5	69
106	130.63	54.10	S-8	200	1			4	3.949	8.9	57.5	12*
107	130.63	54.08	S-9	200	1			4	3.732	3.4	41.2	29*
108	130.57	54.05	S-9	200	1			4	3.592	22.0	48.8	36*

NOTES: Location by longitude (°W) and latitude (°N).
Pluton: B—Banks and Trutch Islands pluton, C—Captain Cove pluton, G—Gil Island complex, and S—Stephens Island pluton.
Phase: 5b—gneissic diorite, 7b—hornblende diorite, 8—quartz diorite, 8a—biotite quartz diorite, 8b—hornblende quartz diorite, 9—granodiorite, 9b—biotite granodiorite, and 10a—biotite quartz monzonite.
Data Screening: (a) core rejected because the 2 specimen directions diverged by more than 25°, (b) core rejected as sole remaining direction to inadequately represent site, and (c) core rejected as anomalous with respect to remaining cores from the site.
R is the length of vector resultant.
α_{95} is the radius of the cone of 95% confidence (Fisher 1953) in degrees.
*is a stable site with $\alpha_{95} < 40°$.
Basic data for the Gil Island complex sites is from Fedikow (1975).

A/m [2]), using a Schonstedt GSD-1 demagnetizer. Using decay plots for intensities and the paleomagnetic stability index (*PSI*) for directions (Symons and Stupavsky 1974), an optimum AF intensity was selected for each site (Table 1) in which the remaining specimens were demagnetized before measuring their cleaned remanent magnetization.

The pilot specimens show the isolation of a single stable component on AF demagnetization. Most specimens exhibit little change in remanence direction between 100 and 1000 Oe (7.96×10^3 and 79.60×10^3 A/m) resulting in low *PSI* values and in a slow exponential decay in intensity (Figs. 2a, b). They have a small initial viscous component which is removed by 100 Oe (7.96×10^3 A/m). A few specimens show minor erratic directional changes on demagnetization giving *PSI* minima in the 300–400 Oe (23.88×10^3–31.84×10^3 A/m) range (Fig. 2c). Generally they show a rapid exponential intensity decay up to 200 Oe (15.92×10^3 A/m) as significant viscous components are removed and random intensity increases above 700 Oe (55.72×10^3 A/m) as anhysteretic components are introduced into the specimens. A few specimens follow a great circle towards a stable end point direction with *PSI* minima in the 550–1000 Oe (43.78×10^3–79.60×10^3 A/m) range (Fig. 2d). These specimens also show an initial rapid exponential intensity decay as viscous components are removed and fluctuations near 1000 Oe (79.60×10^3 A/m) as anhysteretic components are added.

A further 8 representative specimens have been thermally demagnetized at temperature steps of 150, 300, 400, 450, 500, 525, 550, 600, and 650°C using a nonmagnetic induction furnace protected from magnetic fields by mu-metal nested shields in a shielded room. Data for specimens from sites 52, 62, 66, and 98 are shown in Fig. 3. Three specimens (sites 52, 92, and 98) show no initial viscous components whereas five (sites 62, 66, 86, 102, and 104) show such components being removed by 150°C (sites 62 and 104) or 300°C (sites 66, 86, and 102). Seven specimens show the isolation of a single stable component in the same direction as the site mean isolated by AF demagnetization to temperatures above 400°C (sites 92 and 104), 450°C (site 52), 525°C (sites 86, 98, and 102), and 600 or 650°C (site 62). The eighth specimen (site 66) isolates a stable component up to 525°C and swings to a reversed antiparallel position at 550 and 600°C. All specimens at higher temperatures show weak, randomly directed components. These data show that the specimens retain a primary normal remanence in magnetite that is stable through geologic times in the order of 100 Ma (Pullaiah *et al.* 1975). Occasional weak primary hematite magnetization (site 62) or reversed components (site 66) may also be present.

[2] 1 Oe = 79.6 A/m in SI units.

Site Statistics

The stable site mean remanence directions (Table 1) were identified using the following tests. First, if two specimen directions from one core diverged by $>20°$, the core was deemed to be either too weakly or too inhomogeneously magnetized to give a valid core mean direction. Of the original 245 cores, 188 survived this test. Second, in three sites, only one core survived which was rejected as insufficient data to represent the site. Third, in four sites, one core was rejected because its direction deviated by $>60°$ from the site mean direction. Fourth, after combining the core mean directions to give the site mean direction, those sites with a radius of 95% confidence (α_{95}, Fisher 1953) of $>40°$ were rejected as insufficiently well defined. Of the original 49 sites, 36 or 74% passed the tests and are thought to have stable mean directions.

The 13 rejected sites show a marked petrological bias in that 5 are from the 11 gneissic diorite sites and 4 are from the 13 diorite sites. By comparison only 1 of the 10 quartz diorite, 3 of the 12 granodiorite, and none of the 3 monzonite sites were rejected. This bias is opposite to that found in plutons of the eastern and central zones where the more acidic phases proved less stable (Symons 1974*a*, 1977).

Pluton Directions

In the previous paleomagnetic studies of the eastern and central zone plutons (Symons 1974, 1977), the site mean directions were consistent within one phase, within one pluton, and within one zone provided that the structural trend of the zone remained constant. The structural trend in the western zone is constant at N 35° W, so the site directions were combined to calculate the phase mean directions. Examination of the four

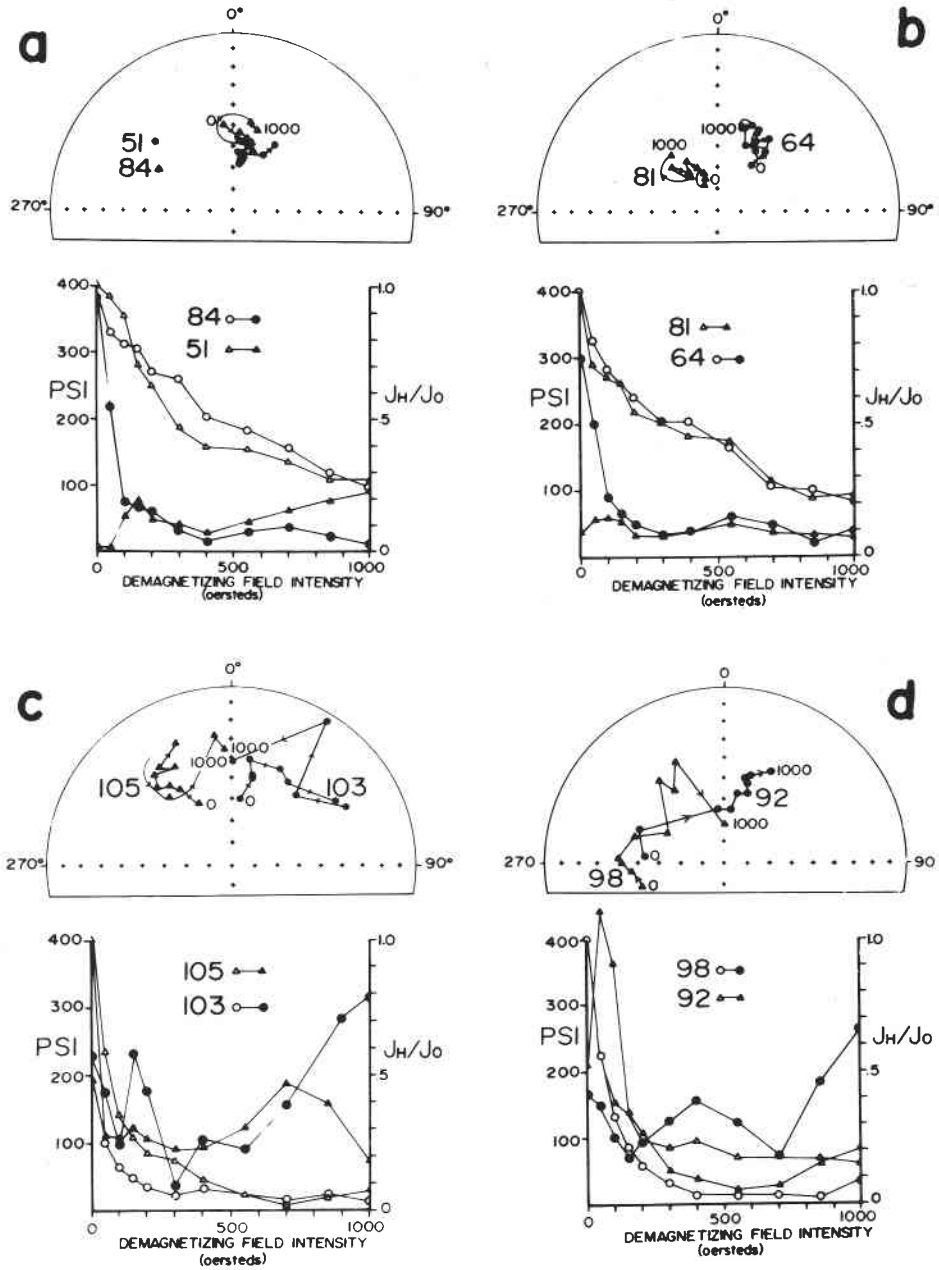


FIG. 2. AF demagnetization data for hornblende diorite (51 and 84), quartz diorite (98 and 103), gneissic diorite (64 and 92), biotite granodiorite (81) and granodiorite (105) specimens. Above is an equal-area stereonet showing the progressive change in remanence direction. Below, plotted against the AF demagnetizing field intensity, is the paleomagnetic stability index (*PSI*) in millidegrees/oersted as solid symbols and the ratio of the demagnetized specimen intensity to original intensity (J_H/J_O).

phase directions for the Captain Cove pluton reveals that they are statistically indistinguishable (Table 2). This is true also for the two Stephens Island phases. Therefore, it is valid to combine the site directions to give the pluton mean directions (Fig. 4, Table 2).

Next, the angular variance ratio test of Larochelle (1969*a*) was used to compare the site

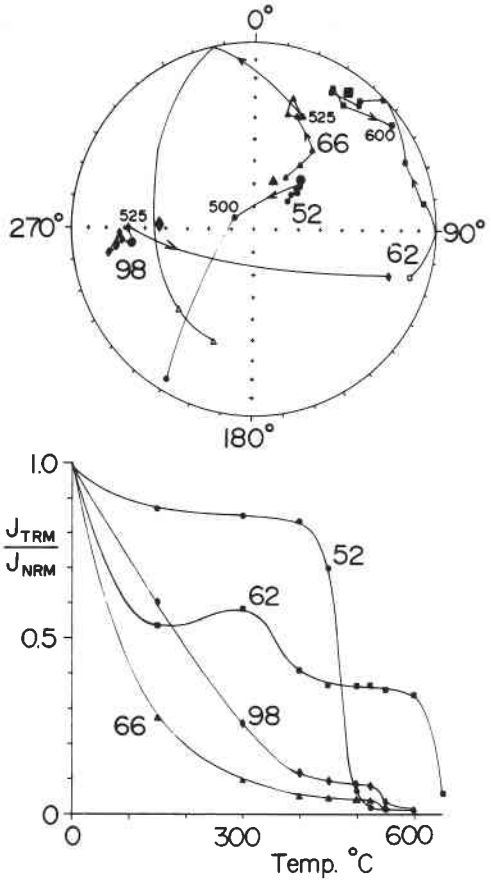


FIG. 3. Thermal demagnetization data for specimens from sites 52 (circles), 62 (squares), 66 (triangles), and 98 (diamond). On the stereonet arrows indicate the direction of increasing temperature with the temperature in °C of some points. The larger symbol indicates the site mean direction from AF demagnetization. The graph shows the decay in the ratio of the demagnetized remanence to original remanence (J_{TRM}/J_{NRM}) with increasing temperature.

mean magnetization directions of the plutons (Table 3). The ratio of the between-unit to within-unit angular variances (δ_b^2/δ_w^2) is much less than the theoretical 95% confidence limit if any pair of the Captain Cove, Stephens Island, and Gil Island direction populations is compared. Therefore the tests are positive meaning that the three pluton directions are not statistically significantly different and they can be validly combined to form one population. Conversely, the Banks Island directions give negative tests and cannot be combined with the other three pluton directions. The eastern and western panels therefore have distinct paleomagnetic direction.

A criticism of the Banks Island tests is that the pluton has only 4 valid sites. For this reason the overall pluton direction was recalculated using all 8 site mean directions regardless of their individual error limits. On this basis the pluton direction changes by an insignificant 4°, the α_{95} value decreases from 32° to 22°, and the variance tests become more significant to greatly exceed the 99% confidence limit. Therefore there is reason to believe that increased sampling of the Banks Island pluton would reinforce the statistical conclusions drawn above.

Remanence Polarity

All valid sites have a normal remanence polarity which is consistent with the known polarity pattern for the Upper Jurassic – Cretaceous (Larson and Hilde 1975) (Fig. 5). The 102 Ma Stephens Island and 109 Ma Captain Cove plutons coincide in age with the 81 to 110 Ma. Cretaceous Quiet Interval (Irving and Pullaiah 1976). The 136 Ma Gil Island and 144 Ma Banks Island plutons coincide with the top of the predominantly normal 140 to 165 Ma Upper Jurassic Quiet Interval but they may be normal by simple chance. Nevertheless the substantial agreement of the predictable and observed polarities suggests that the plutons preserve a primary remanence acquired when the K–Ar ages were frozen.

Pole Positions

Pole positions calculated from the four plutons are listed in Table 4 along with Jurassic and Cretaceous reference poles for North America obtained chiefly from rock units in the stable craton.

The eastern panel poles for the Stephens Island, Captain Cove, and Gil Island plutons differ significantly from the reference poles (Fig. 6, Table 4). The simplest explanation is that the panel has been tilted over to the southwest by $\approx 30°$ about a horizontal axis parallel to the N 35° W plutonic trend. Two structural features favour this hypothesis. Assuming that the plutons were emplaced with vertical contacts, they are now tilted $\approx 30°$ from the vertical in the correct sense (Fig. 7a). In addition the foliation attitudes in the panel show the same tilt bias (Fig. 7b). Rotation about a vertical axis is unlikely because the plutonic trend is constant at N 35° W throughout the panel and is parallel to the overall Coast plutonic complex trend. Two paleomagnetic arguments also favour this hypo-

TABLE 2. Mean remanence directions for groups of sites

Group	Sites	No. of sites	Mean remanence direction			
			Length (R)	Decl.	Incl. (+ down)	α_{95}
Captain Cove pluton						
gneissic diorite	60, 65, 69, 82, 83	5	4.861	15	51	15
quartz diorite	57, 58, 66, 70	4	3.903	23	70	17
granodiorite	67, 68, 80, 81	4	3.912	20	61	16
monzonite	85, 86, 87	3	2.947	38	64	20
overall	all above	16	15.478	21	61	7.1
Stephens Island pluton						
quartz diorite	101–103, 106	4	3.971	12	58	9.1
granodiorite	104, 107, 108	3	2.981	14	46	12
overall	101–104, 106–108	7	6.917	13	53	7.2
Stephens Island and Captain Cove plutons	all above	23	22.310	18	58	5
Gil Island diorite complex	50, 52, 62, 63, 84, 88–90	9	8.190	29	50	17
Eastern panel plutons	all above	32	30.399	22	56	5.9
Banks Island pluton						
gneissic diorite	92	1		2	65	
quartz diorite	95	1		348	83	
granodiorite	98, 100	2	1.724	313	70	
overall	92, 95, 98, 100	4	3.671	335	73	32
(overall*)	92, 93, 95–100	8	7.028	345	70	22

NOTES: R is the vector resultant. The declination, inclination, and radius of 95% confidence (α_{95}) are in degrees. The asterisk indicates unscreened data.

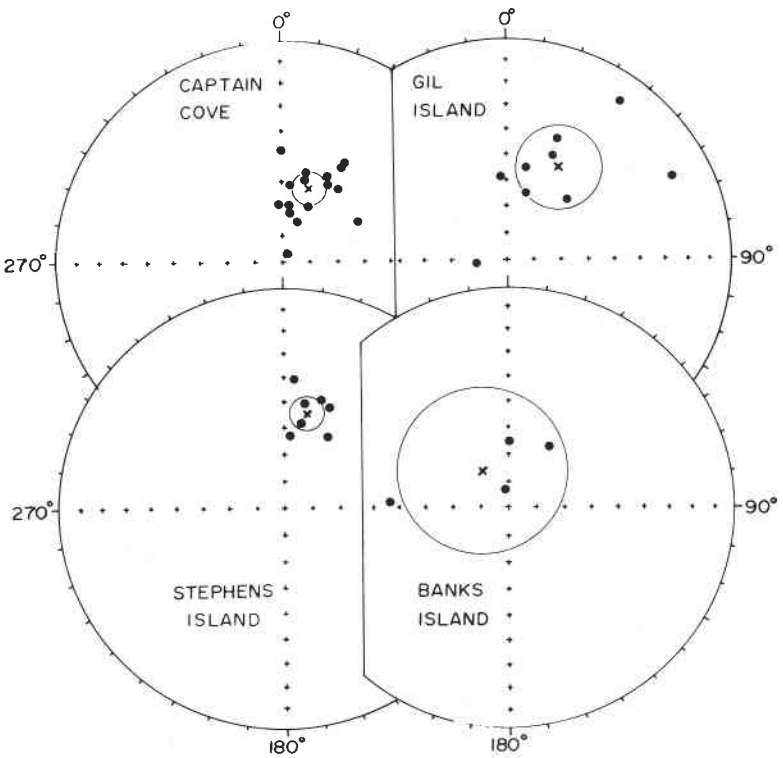


FIG. 4. Equal-area stereonets showing the site mean directions in the lower hemisphere for each pluton along with its mean position (X) and cone of 95% confidence.

TABLE 3. Angular variance ratio analysis between plutons

Test units compared	Number of sites (B)	Variance ratio δ_b^2/δ_w^2	Theoretical statistics $F_{0.05}$	Correlation test results
Captain Cove vs. Stephens Island	23	2.16	3.23	Positive
Captain Cove vs. Gil Island	25	1.89	3.20	Positive
Stephens Island vs. Gil Island	16	1.04	3.34	Positive
Banks Island vs. Captain Cove	20	4.18	3.26	Negative
Banks Island vs. Stephens Island	11	5.37	3.55	Negative
Banks Island vs. Gil Island	13	3.98	3.44	Negative
Banks Island vs. Eastern panel	36	5.51	3.13	Negative

NOTES: The variance ratio δ_b^2/δ_w^2 is the between-unit over the within-unit angular variances in (radians)^2. It is compared with the 95% confidence limit. $F_{0.05} = F_{2,2(B-2),0.05}$.

TABLE 4. Pole positions

Rock unit	Age (Ma)	Pole position °W	°N	°δp	°δm	References
Stephens Island pluton	102± 8	339	67	7	10	this study
Captain Cove pluton	109± 6	9	72	8	11	this study
Stephens Island and Captain Cove	105±10	357	70	6	8	this study
Gil Island complex	136± 3	4	59	15	23	this study
Eastern panel		360	67	6	9	this study
Banks Island pluton	144± 6	185	75	52	58	this study
Banks Island unscreened		210	81	33	38	this study
*Cretaceous reference**						
Niobara limestone	79±10	167	62			Shive and Frerichs (1974)
Elkhorn Mtn. volcanics	80	171	69			Hanna (1967)
Sierra Nevada pluton	86± 4	165	69			Grommé and Merrill (1965)
Arkansas alkalic complex	97±10	170	70			Scharon and Hsu (1969)
Isachsen diabase	106± 5	180	69			Larochelle and Black (1963)
Monteregian Hills plugs	111±10	171	71			Larochelle (1969), Larochelle *et al.* (1965)
Mt. Megantic complex	113	189	68			Larochelle (1969), Fairbairn *et al.* (1963)
Mt. Ascutney gabbro	130± 5	173	64			Opdyke and Wensink (1966), Foland *et al.* (1971)
Cretaceous average		173	68	3	3	
*Jurassic reference**						
Topley monzonites	139± 4	231	70			Symons (1973*d*)
Morrison sandstones	142± 8	205	65			Steiner and Helsley (1975)
Summerville sandstones	149±13	274	70			Steiner and Helsley (1972)
Belknap-Monadock stocks	158± 4	331	89			Opdyke and Wensink (1966), Foland *et al.* (1971)
Appalachian dikes	170±15	258	65			Smith (1976)
White Mtn. batholith	185± 5	191	76			Opdyke and Wensink (1966), Foland *et al.* (1971)
Jurassic average		234	75	11	11	

*Pole positions from deformed rock units in the western Cordillera are excluded from the North American cratonic Reference Pole.

thesis. As shown in Fig. 8, the eastern panel appears to preserve a tilted segment of the Upper Jurassic to mid Cretaceous polar wander path with the 136 Ma Gil Island pole returning on tectonic correction close to the Upper Jurassic position and the 106 Ma Stephens Island – Captain Cove pole returning close to the mid Cretaceous position. This suggests that the panel was tilted after emplacement of the 102 Ma Stephens Island pluton. Symons (1974*a*, 1977) shows paleomagnetically that the plutonic rocks in the adjacent 64–80 Ma central zone to the east

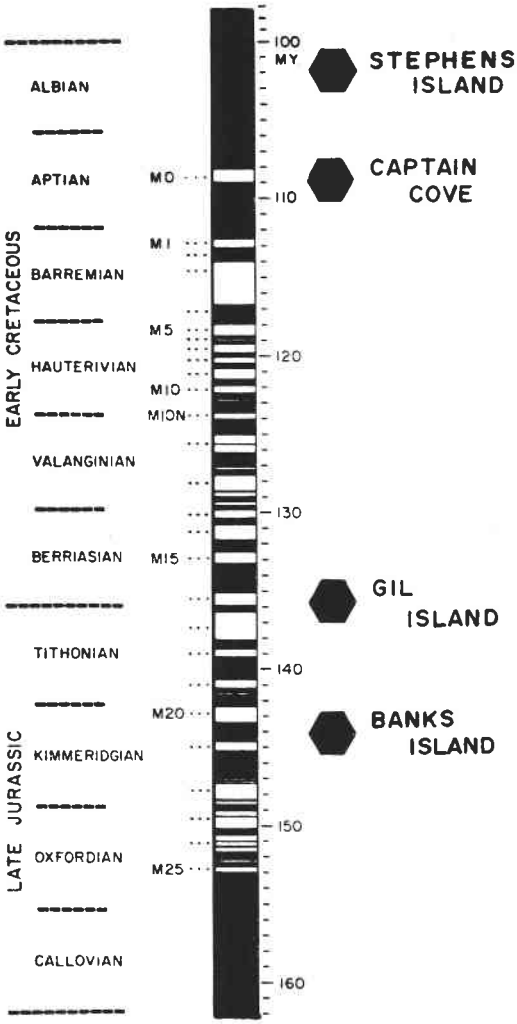


FIG. 5. Polarity time scale for mid Jurassic to mid Cretaceous time after Larson and Hilde (1975). Normal intervals are solid and reversed intervals open. The K–Ar age spans for the four sampled plutons are on the right.

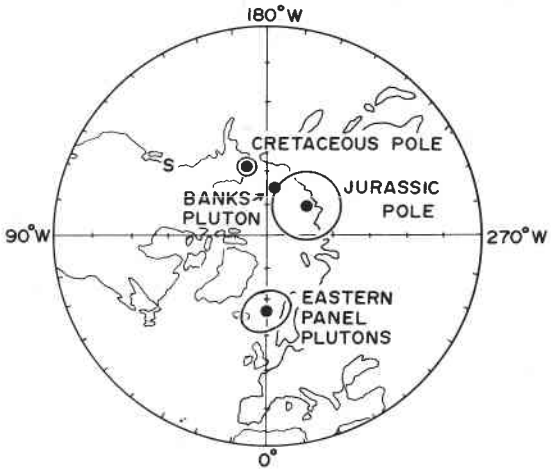


FIG. 6. Polar projection of the northern hemisphere showing the eastern (Stephens Island, Captain Cove, and Gil Island plutons) and western (Bank Island) panel poles along with the Jurassic and Cretaceous reference poles (Table 4). Also shown are the ovals of 95% confidence.

were tilted in the same sense $\approx 15°$ SW before the Eocene, and from a fold test that their remanence predates the post-Eocene Hawkesbury Warp deformation. Therefore the eastern panel must have been tilted during Upper Cretaceous or Paleocene time.

The western panel Banks Island pole is concordant with the Jurassic reference pole (Fig. 6) suggesting that the panel has not been tilted. Conversely, the few available foliation attitudes (Fig. 7c) favour a $\approx 22°$ SW tilt. Unfortunately, there are insufficient paleomagnetic data to resolve this conflict because the error limits about the pole exceed the possible tilt.

The basic conclusion drawn from the AF demagnetization data, the remanence polarity data, and the pole data is that these plutonic rocks retain a primary thermoremanence acquired during emplacement. The remanence is contemporaneous with the K–Ar dates, and it has survived all subsequent deformation episodes including uplift, tilt, and folding.

Geotectonics

About two-thirds of the paleomagnetic poles from Cordilleran rock units are discordant. A possible explanation is that the earth's magnetic field underwent periodic excursions from its usual position (Grommé and Gluskoter 1965; Saad 1969), however, few Mesozoic or Cenozoic rock units from the craton record possible excursions. A second explanation is that the western Cordillera was translated northward by some 30° to 60° relative to the craton on a San Andreas type transform fault (Beck and Noson 1972; Irving and Yole 1972). From mid Jurassic onwards this explanation is incompatible with the finding of concordant Jurassic and younger poles in the Insular Belt (Symons 1971*a*, *b*, 1973*a*), Coast plutonic complex (this study; Symons 1973*b*), and Interior Plateau (Symons 1973*d*) as

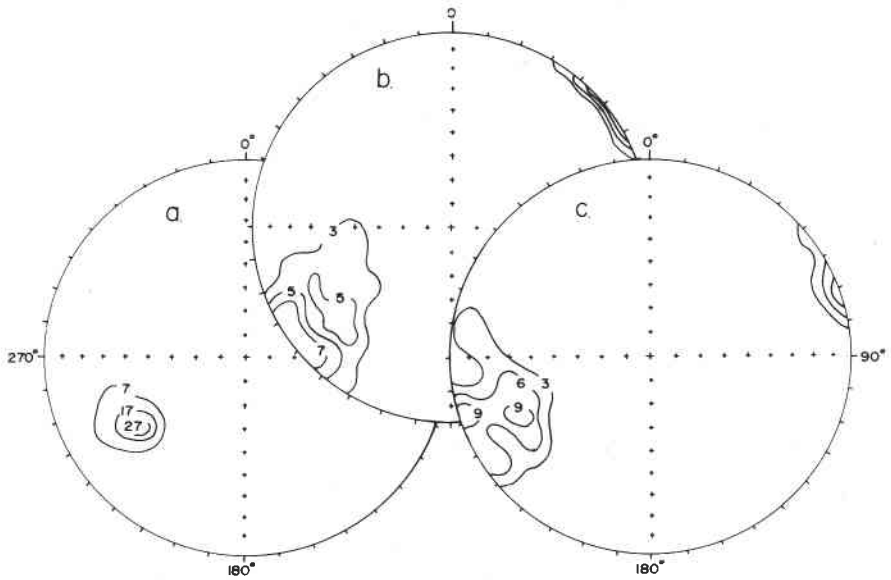


FIG. 7. Equal-area stereonet showing percent density contours of poles for: (a) 11 eastern panel plutonic contacts; (b) 154 eastern panel plutonic foliations; and (c) 51 western panel plutonic foliations. Data from Hutchison (1967) and Roddick (1970).

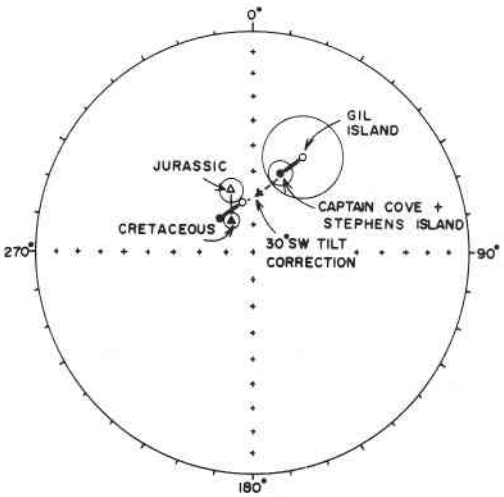


FIG. 8. Equal-area stereonet showing the eastern panel mid Cretaceous (Stephens Island and Captain Cove plutons) and upper Jurassic (Gil Island complex) remanence directions and their movement upon 30° SW tilt correction to beside the corresponding reference directions.

well as with near-sided poles (Grommé *et al.* 1967). A third explanation combines northward translation with variable clockwise rotation of tectonic blocks at the leading edge of the North American plate (Teissere and Beck 1973; Beck 1975). This model fails to explain the concordant and near-sided poles and in addition poles showing counter-clockwise rotation (Symons 1977). Conversely, most concordant and discordant poles can be rationally explained in terms of regional tectonics (Watkins 1965; Symons 1971*c*, 1973*c*, 1974*b*: Whitney 1975) although a few are difficult to explain in this way (Cox 1957; Packer and Stone 1972).

The pole positions for the four plutons in this study can also be explained best by regional tectonics with the western panel untilted and the eastern panel tilted $\approx 30°$ SW. This tilting occurred during orogenic activity in Upper Cretaceous or Paleocene time. The westward-moving North American continental plate overrode the oceanic Kula plate (Atwater 1970) to form an easterly-dipping subduction zone (Monger *et al.* 1972; Hyndman 1976). Plutonic emplacement moved sequentially eastward to form the western panel of the western zone (140–150 Ma), the eastern panel (80–115 Ma), the central zone (64–80 Ma), and the eastern zone (40–50 Ma). During subduction, the rocks of the eastern panel and central zone were tilted $\approx 30°$ and 20° SW respectively whereas those of the outer western panel and eastern zone were not tilted. Post-Eocene deformation led to formation of the Hawkesbury Warp (Symons 1977) and to the varying amounts of uplift on the tectonic panels and zones.

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